Ex. No. 7	Application State Management and Serialization
Date of Exercise	12/10/26
GitHub Host Link	https://github.com/jomiwilson/flutter_lab_exp7

Aim

To manage application state and serialize/deserialize data using JSON.

Description

This experiment introduces application state management and JSON serialization in Flutter. A Student model class is created with toJson and fromJson methods to convert between Dart objects and JSON. A list is maintained as application state and updated using setState, and the data is displayed using ListView.builder.

Tools/Software Required: Flutter

Procedure:

Step 1

Create project:

```
flutter create state_app
cd state_app
code .
```

Step 2 – Create Student model

Create:

```
lib/student.dart
```

Add:

```
class Student {
  final int id;
  final String name;
  final String course;
  Student({
    required this.id,
    required this.name,
    required this.course,
  });
  Map<String, dynamic> toJson() {
    return {
      'id': id,
      'name': name,
      'course': course,
    };
  }
```

```
  factory Student.fromJson(Map<String, dynamic> json) {
    return Student(
      id: json['id'],
      name: json['name'],
      course: json['course'],
    );
  }
}
```

Step 3 – Create student object

```
Student student = Student(
  id: 1,
  name: 'Viknan S',
  course: 'CSE',
);
```

Step 4 – Convert object to JSON

```
Map<String, dynamic> data = student.toJson();
print(data);
```

Step 5 – Convert JSON back to object

```
Student newStudent = Student.fromJson(data);
print(newStudent.name);
```

Step 6 – Add state

Create a list:

```
List<String> students = [];
```

Add student:

```
setState(() {
  students.add('Viknan S');
});
```

Step 7 – Display list

```
ListView.builder(
itemCount: students.length,
  itemBuilder: (context, index) {
    return ListTile(
      title: Text(students[index]),
    );
  },
)
```

Program

lib/student.dart

```
class Student {
  final int id;
  final String name;
  final String course;

  Student({
```

```
    required this.id,
    required this.name,
    required this.course,
  });

  // Convert a Student object into JSON (Map)
  Map<String, dynamic> toJson() {
    return {
      'id': id,
      'name': name,
      'course': course,
    };
  }

  // Create a Student object from JSON (Map)
  factory Student.fromJson(Map<String, dynamic> json) {
    return Student(
      id: json['id'],
      name: json['name'],
      course: json['course'],
    );
  }
}
```

lib/main.dart

```
import 'dart:convert';
import 'package:flutter/material.dart';
import 'student.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return const MaterialApp(
      debugShowCheckedModeBanner: false,
      home: StudentPage(),
    );
  }
}

class StudentPage extends StatefulWidget {
  const StudentPage({super.key});

  @override
  State<StudentPage> createState() => _StudentPageState();
}
```

```
class _StudentPageState extends State<StudentPage> {
  // Application state: list of students
  final List<Student> students = [];
  final TextEditingController nameController = TextEditingController();
  String lastJson = '';

  void addStudent() {
    if (nameController.text.isEmpty) return;

    Student student = Student(
      id: students.length + 1,
      name: nameController.text,
      course: 'CSE',
    );

    // Serialization: object -> JSON
    Map<String, dynamic> data = student.toJson();
    String jsonString = jsonEncode(data);
    print(jsonString);

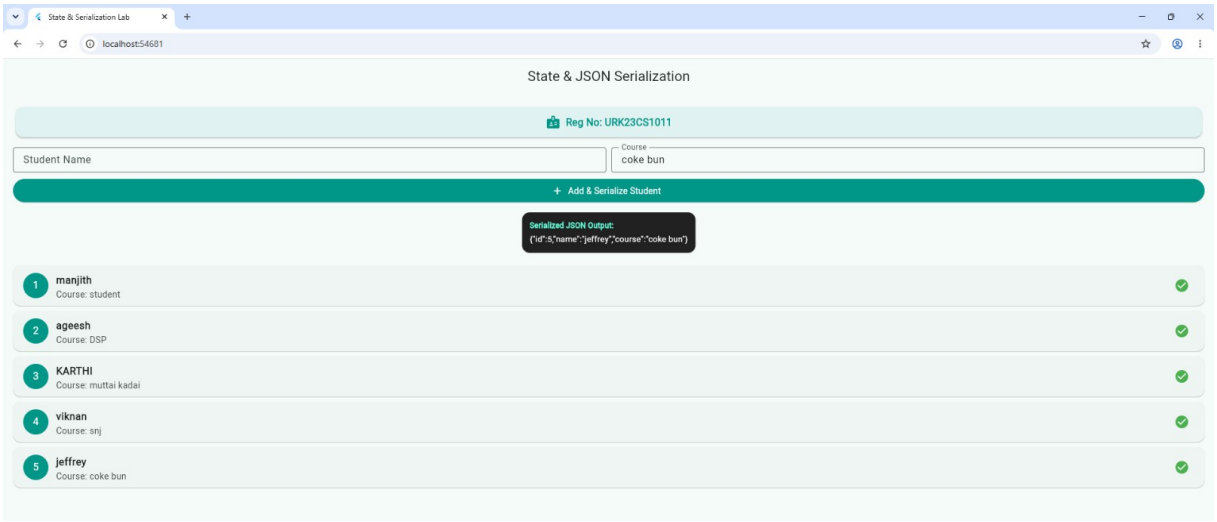
    // Deserialization: JSON -> object
    Student newStudent = Student.fromJson(jsonDecode(jsonString));
    print(newStudent.name);

    setState(() {
      students.add(newStudent);
      lastJson = jsonString;
    });
    nameController.clear();
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('State & JSON Serialization'),
        centerTitle: true,
      ),
      body: Padding(
        padding: const EdgeInsets.all(16),
        child: Column(
          children: [
            const Text(
              'Reg No: URK23CS1299',
              style: TextStyle(fontWeight: FontWeight.bold, color:
Colors.teal),
            ),
            const SizedBox(height: 12),
            TextField(
              controller: nameController,
              decoration: const InputDecoration(
                labelText: 'Student Name',
                border: OutlineInputBorder(),
```

```
              ),
            ),
            const SizedBox(height: 12),
            SizedBox(
              width: double.infinity,
              child: ElevatedButton(
                onPressed: addStudent,
                child: const Text('Add Student'),
              ),
            ),
            const SizedBox(height: 8),
            if (lastJson.isNotEmpty) Text('JSON: $lastJson'),
            const SizedBox(height: 8),
            Expanded(
              child: ListView.builder(
                itemCount: students.length,
                itemBuilder: (context, index) {
                  return ListTile(
                    leading: CircleAvatar(child: Text('$
{students[index].id}')),
                    title: Text(students[index].name),
                    subtitle: Text(students[index].course),
                  );
                },
              ),
            ),
          ],
        ),
      ),
    );
  }
}
```

Output



Result

A Flutter application for state management and serialization was developed successfully. The Student object was converted to JSON using toJson and converted back using fromJson. The student list was updated using setState and displayed dynamically using ListView.builder. The application maintained changing student data and converted data between Dart objects and JSON.